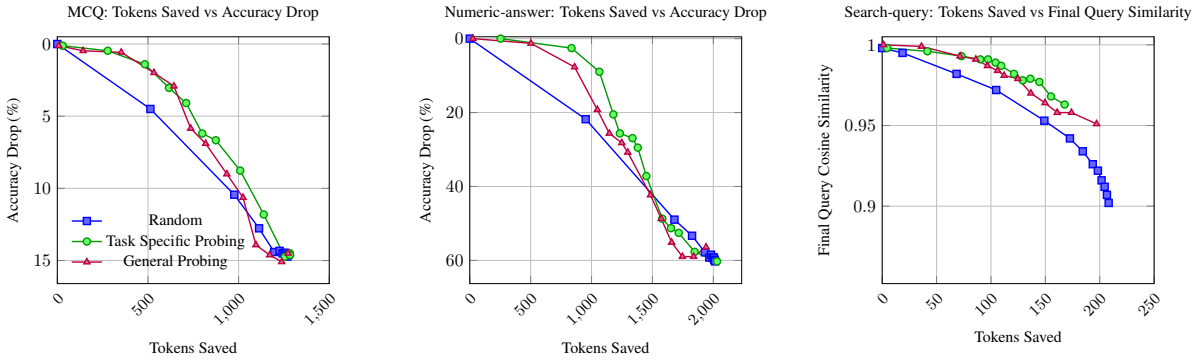


Figure 4. Early stopping performance for random, task-specific and generic probe-based gates on Qwen3-4B reasoning.

Despite its promising results, our study is limited to the models, tasks, and reasoning lengths evaluated in this work. We do not study substantially larger models or much longer reasoning traces, where answer dynamics may differ. In addition, our experiments focus on language based reasoning tasks, and we leave extensions to broader multimodal models for future work.

7. Impact Statement

This work presents an inference-time approach for reducing unnecessary reasoning generation in large language models. By stopping generation once the answer has stabilized, the method lowers token usage, inference cost, and latency without modifying the base model. These efficiency gains can help reduce the computational and environmental footprint of deploying reasoning models at scale.

References

Balunović, M., Dekoninck, J., Petrov, I., Jovanović, N., and Vechev, M. Matharena: Evaluating llms on uncontaminated math competitions, February 2025. URL https://matharena.ai/.

Bogdan, P. C., Macar, U., Nanda, N., and Conmy, A. Thought anchors: Which llm reasoning steps matter? *arXiv preprint arXiv:2506.19143*, 2025.

Center for AI Safety, Scale AI, and HLE Contributors Consortium. A benchmark of expert-level academic questions to assess AI capabilities. *Nature*, 649:1139–1146, 2026. doi: 10.1038/s41586-025-09962-4. URL https://arxiv.org/abs/2501.14249.

Karpas, E., Abend, O., Belinkov, Y., Lenz, B., Lieber, O., Ratner, N., Shoham, Y., Bata, H., Levine, Y., Leyton-Brown, K., et al. Mrkl systems: A modular, neuro-symbolic architecture that combines large language models, external knowledge sources and discrete reasoning. *arXiv preprint arXiv:2205.00445*, 2022.

Kojima, T., Gu, S. S., Reid, M., Matsuo, Y., and Iwasawa, Y. Large language models are zero-shot reasoners. *Advances in neural information processing systems*, 35: 22199–22213, 2022.

Lampinen, A., Dasgupta, I., Chan, S., Mathewson, K., Tessler, M., Creswell, A., McClelland, J., Wang, J., and Hill, F. Can language models learn from explanations in context? In *Findings of the Association for Computational Linguistics: EMNLP 2022*, pp. 537–563, 2022.

Lanham, T., Chen, A., Radhakrishnan, A., Steiner, B., Denison, C., Hernandez, D., Li, D., Durmus, E., Hubinger, E., Kernion, J., Lukošiūtė, K., Nguyen, K., Cheng, N., Joseph, N., Schiefer, N., Rausch, O., Larson, R., McCandlish, S., Kundu, S., Kadavath, S., Yang, S., Henighan, T., Maxwell, T., Telleen-Lawton, T., Hume, T., Hatfield-Dodds, Z., Kaplan, J., Brauner, J., Bowman, S. R., and Perez, E. Measuring faithfulness in chain-of-thought reasoning, 2023a. URL https://arxiv.org/abs/2307.13702.

Lanham, T., Chen, A., Radhakrishnan, A., Steiner, B., Denison, C., Hernandez, D., Li, D., Durmus, E., Hubinger, E., Kernion, J., et al. Measuring faithfulness in chain-of-thought reasoning. *arXiv preprint arXiv:2307.13702*, 2023b.

Lewkowycz, A., Andreassen, A., Dohan, D., Dyer, E., Michalewski, H., Ramasesh, V., Slone, A., Anil, C., Schlag, I., Gutman-Solo, T., et al. Solving quantitative reasoning problems with language models. *Advances in neural information processing systems*, 35:3843–3857, 2022.

Liang, X., Song, S., Zheng, Z., Wang, H., Yu, Q., Li, X., Li, R.-H., Wang, Y., Wang, Z., Xiong, F., et al. Internal consistency and self-feedback in large language models: A survey. *arXiv preprint arXiv:2407.14507*, 2024.

Lightman, H., Kosaraju, V., Burda, Y., Edwards, H., Baker, B., Lee, T., Leike, J., Schulman, J., Sutskever, I., and

Cobbe, K. Let's verify step by step. *arXiv preprint arXiv:2305.20050*, 2023.

Mao, Z. and Venkat, A. Recurrent confidence chain: Temporal-aware uncertainty quantification in large language models. *arXiv preprint arXiv:2601.13368*, 2026.

Mao, Z., Venkat, A., Bisliouk, A., Kothiyal, A., Subramanian, S. K., Singhu, S., and Ruchkin, I. Confidence over time: Confidence calibration with temporal logic for large language model reasoning. *arXiv preprint arXiv:2601.13387*, 2026.

Mei, Z., Zhang, C., Yin, T., Lidard, J., Shorinwa, O., and Majumdar, A. Reasoning about uncertainty: Do reasoning models know when they don't know? *arXiv preprint arXiv:2506.18183*, 2025.

Monteiro, J., Noel, P.-A., Marcotte, E., Rajeswar, S., Zantedeschi, V., Vazquez, D., Chapados, N., Pal, C., and Taslakian, P. Repliqa: A question-answering dataset for benchmarking llms on unseen reference content, 2024. URL https://arxiv.org/abs/2406.11811.

Plaat, A., Wong, A., Verberne, S., Broekens, J., Van Stein, N., and Bäck, T. Multi-step reasoning with large language models, a survey. *ACM Computing Surveys*, 58(6):1–35, 2025.

QuotientAI. Mcp tool call evaluation test dataset, 2025. URL https://huggingface.co/datasets/quotientai/limbic-eval-tool-use-mcp.

Reimers, N. and Gurevych, I. all-minilm-l6-v2: Sentence embeddings model. https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2, 2019. Hugging Face model card, accessed January 27, 2026.

Rein, D., Hou, B. L., Stickland, A. C., Petty, J., Pang, R. Y., Dirani, J., Michael, J., and Bowman, S. R. Gpqa: A graduate-level google-proof q&a benchmark, 2023. URL https://arxiv.org/abs/2311.12022.

Shao, Z., Wang, P., Zhu, Q., Xu, R., Song, J., Bi, X., Zhang, H., Zhang, M., Li, Y., Wu, Y., et al. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.

Wan, A., Wallace, E., and Klein, D. What evidence do language models find convincing? *arXiv preprint arXiv:2402.11782*, 2024.

Wang, X., Wei, J., Schuurmans, D., Le, Q., Chi, E., Narang, S., Chowdhery, A., and Zhou, D. Self-consistency improves chain of thought reasoning in language models. *arXiv preprint arXiv:2203.11171*, 2022.

Wang, Y., Ma, X., Zhang, G., Ni, Y., Chandra, A., Guo, S., Ren, W., Arulraj, A., He, X., Jiang, Z., Li, T., Ku, M., Wang, K., Zhuang, A., Fan, R., Yue, X., and Chen, W. Mmlu-pro: A more robust and challenging multi-task language understanding benchmark, 2024. URL https://arxiv.org/abs/2406.01574.

Wei, J., Wang, X., Schuurmans, D., Bosma, M., Xia, F., Chi, E., Le, Q. V., Zhou, D., et al. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems*, 35:24824–24837, 2022.

Wiegreffe, S. and Pinter, Y. Attention is not not explanation. *arXiv preprint arXiv:1908.04626*, 2019.

Yang, A., Li, A., Yang, B., Zhang, B., Hui, B., Zheng, B., Yu, B., Gao, C., Huang, C., Lv, C., Zheng, C., Liu, D., Zhou, F., Huang, F., Hu, F., Ge, H., Wei, H., Lin, H., Tang, J., Yang, J., Tu, J., Zhang, J., Yang, J., Yang, J., Zhou, J., Zhou, J., Lin, J., Dang, K., Bao, K., Yang, K., Yu, L., Deng, L., Li, M., Xue, M., Li, M., Zhang, P., Wang, P., Zhu, Q., Men, R., Gao, R., Liu, S., Luo, S., Li, T., Tang, T., Yin, W., Ren, X., Wang, X., Zhang, X., Ren, X., Fan, Y., Su, Y., Zhang, Y., Zhang, Y., Wan, Y., Liu, Y., Wang, Z., Cui, Z., Zhang, Z., Zhou, Z., and Qiu, Z. Qwen3 technical report, 2025. URL https://arxiv.org/abs/2505.09388.

Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K. R., and Cao, Y. React: Synergizing reasoning and acting in language models. In *The eleventh international conference on learning representations*, 2022.

Yoon, D., Kim, S., Yang, S., Kim, S., Kim, S., Kim, Y., Choi, E., Kim, Y., and Seo, M. Reasoning models better express their confidence. *arXiv preprint arXiv:2505.14489*, 2025.

Zhang, B. and Zhang, R. Cot-uq: Improving response-wise uncertainty quantification in llms with chain-of-thought. *arXiv preprint arXiv:2502.17214*, 2025.

Zhou, D., Schärli, N., Hou, L., Wei, J., Scales, N., Wang, X., Schuurmans, D., Cui, C., Bousquet, O., Le, Q. V., and Chi, E. H. Least-to-most prompting enables complex reasoning in large language models. In *The Eleventh International Conference on Learning Representations*, 2023. URL https://openreview.net/forum?id=WZH7099tgfM.

A. Datasets

We use the sample sizes described in Table 2 to conduct our analyses and early stopping experiments for all the datasets. We shuffle and randomly sample from the dataset to obtain samples.

Dataset	S
MCQ	1000
Numeric-answer	500
Search-query	1000
Tool-selection	1000

Table 2. Sample Sizes of reasoning traces

B. Answer Prefix

We choose an answer prefix P that elicits a probability distribution over the task answer space from the model's vocabulary-level output distribution. The choice of P depends on the task format. Below, we describe the prefixes used for each class of task considered in this work.

B.1. MCQ Answer Prefix

For multiple-choice questions (MCQs), we instruct the model to respond in a fixed JSON format: {"answer": "A"}, where A can be replaced by any valid answer label. During forced decoding, we set the answer prefix to $P =$ {"answer": ", which induces a distribution over the possible MCQ answer labels at the next token position. We intervene at each reasoning step and record the unnormalized logits corresponding to the candidate answer labels.

Component	Value
Expected output format	{"answer": "A"}
Answer prefix P	{"answer": "
Predicted token	A, B, C or D

Table 3. Answer prefix used for MCQ tasks.

B.2. Numeric Answer Prefix

For tasks whose answers are numeric, we instruct the model to output its answer inside \boxed{} tags, e.g., \boxed{42}. During forced decoding, we set $P =$ \boxed{ to elicit a distribution over possible numeric answers. We restrict our analysis to questions whose correct answer is represented by a single token in the model vocabulary.

Component	Value
Expected output format	\boxed{42}
Answer prefix P	\boxed{
Predicted token	Single-token numeric answer

Table 4. Answer prefix used for numeric-answer tasks.

B.3. Tool Selection Prefix

For user requests where the model must select exactly one tool from a provided list, we track the evolution of the selected tool across reasoning steps. To simplify the setup, we consider tools without arguments. We use the model's native tool-calling format and choose a prefix such that the next generated token corresponds to the tool name.

For models using Hermes-style function calling (e.g., Qwen3), the format is:

Prefix Format
C<think>$R_1R_2...R_k$</think> <tool_call>{"name": "

Table 5. Answer prefix used for tool selection tasks. The highlighted token corresponds to the predicted tool name.

B.4. Tool Argument Prefix

For tasks where the model is forced to call a specific tool that takes a single argument (e.g., a web search tool), we track the evolution of the generated argument across reasoning steps. Generation proceeds using the same sampling settings and is terminated at the closing tool call tag (</tool_call>).

The prefix used to elicit the argument token is shown below.

Prefix Format
C<think>$R_1R_2...R_k$</think> <tool_call>{"name": "web_search", "arguments": {"query": "

Table 6. Answer prefix used for tool argument generation. The highlighted token corresponds to the generated argument.

C. Full Early Stopping Results

We evaluate early stopping methods on Qwen/Qwen3-4B and Qwen/Qwen3-8B.

C.1. Qwen/Qwen3-4B

Tables 7-15 show the results of early stopping experiments for Qwen/Qwen3-4B

C.1.1. RANDOM PROBING

Table 7. MCQ

Accuracy drop	Tokens saved	Token %
14.620	1273	99.2
14.596	1271	99.0
14.725	1268	98.8
14.678	1264	98.5
14.480	1259	98.1
14.468	1253	97.6
14.515	1243	96.8
14.327	1226	95.5
14.409	1196	93.2
12.772	1113	86.7
10.444	976	76.1
4.503	513	40.0
0.000	0	0.0

Table 8. Numeric-answer

Accuracy drop	Tokens saved	Token %
60.256	2014	99.2
59.872	2012	99.1
59.872	2008	98.9
59.231	2005	98.7
59.359	2000	98.5
59.231	1993	98.2
58.462	1981	97.6
59.231	1966	96.8
57.821	1932	95.2
53.333	1825	89.9
48.974	1681	82.8
21.795	952	46.9
0.000	0	0.0

Table 9. Search-query

Cosine similarity	Tokens saved	Token %
0.902	208	91.0
0.907	207	90.3
0.912	205	89.4
0.916	202	88.2
0.922	198	86.7
0.926	194	84.7
0.934	184	80.6
0.942	173	75.5
0.953	149	65.2
0.972	105	45.8
0.982	68	29.9
0.995	19	8.2
0.998	0	0.0

C.1.2. TASK-SPECIFIC PROBING

Table 10. MCQ

Accuracy drop	Tokens saved	Token %
14.620	1283	100.0
14.620	1283	100.0
14.503	1283	99.9
14.737	1255	97.8
11.813	1139	88.7
8.772	1009	78.6
6.667	874	68.1
6.199	800	62.3
4.094	711	55.4
3.041	616	48.0
1.404	483	37.6
0.468	278	21.7
0.117	30	2.3

Table 11. Numeric-answer

Accuracy drop	Tokens saved	Token %
60.256	2030	100.0
57.692	1846	90.9
52.564	1716	84.5
51.282	1652	81.4
48.718	1581	77.9
37.179	1450	71.4
29.487	1380	68.0
26.923	1336	65.8
25.641	1233	60.7
20.513	1179	58.1
8.974	1064	52.4
2.564	836	41.2
0.000	254	12.5

Table 12. Search-query

Cosine similarity	Tokens saved	Token %
0.963	168	73.3
0.968	155	67.9
0.977	144	63.2
0.979	136	59.6
0.978	129	56.5
0.982	121	53.0
0.987	109	47.8
0.989	104	45.6
0.991	97	42.6
0.991	90	39.4
0.993	73	31.8
0.996	42	18.2
0.998	5	2.0